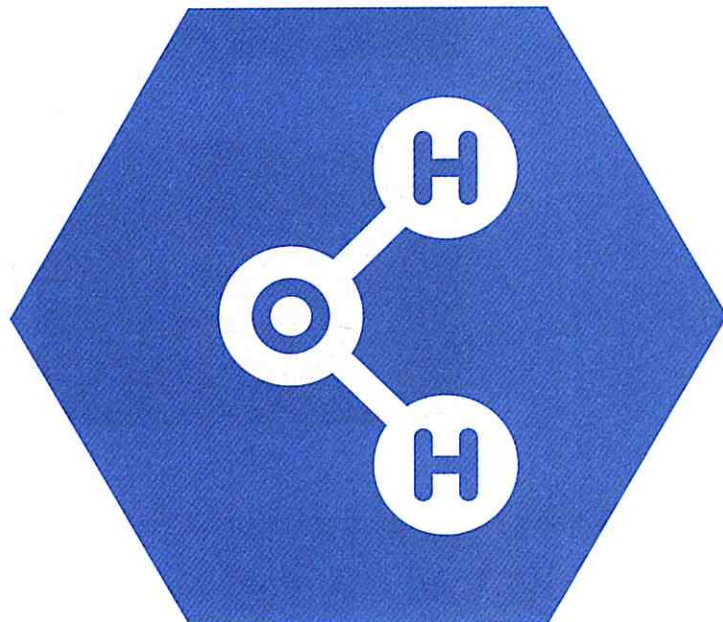




UNSW Global
AUSTRALIA

PAPER B



2015 ICAS

International Competitions
and Assessments for Schools

SCIENCE

DO NOT OPEN THIS BOOKLET UNTIL INSTRUCTED.

30 QUESTIONS

TIME ALLOWED: 45 MINUTES

STUDENT'S NAME:

Read the instructions on the **ANSWER SHEET** and fill in your **NAME, SCHOOL** and **OTHER INFORMATION**.

Use a 2B or B pencil.

Do **NOT** use a pen.

Rub out any mistakes completely.

You **MUST** record your answers on the **ANSWER SHEET**.

Mark only **ONE** answer for each question.

Your score will be the number of correct answers.

Marks are **NOT** deducted for incorrect answers.

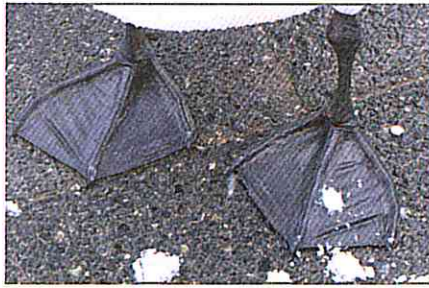
There are **30 MULTIPLE-CHOICE QUESTIONS** (1–30).

Use the information provided to choose the **BEST** answer from the four possible options.

On your **ANSWER SHEET** fill in the oval that matches your answer.

You may use a calculator and a ruler.

1. These photographs show the feet of four different birds.



1



2



3

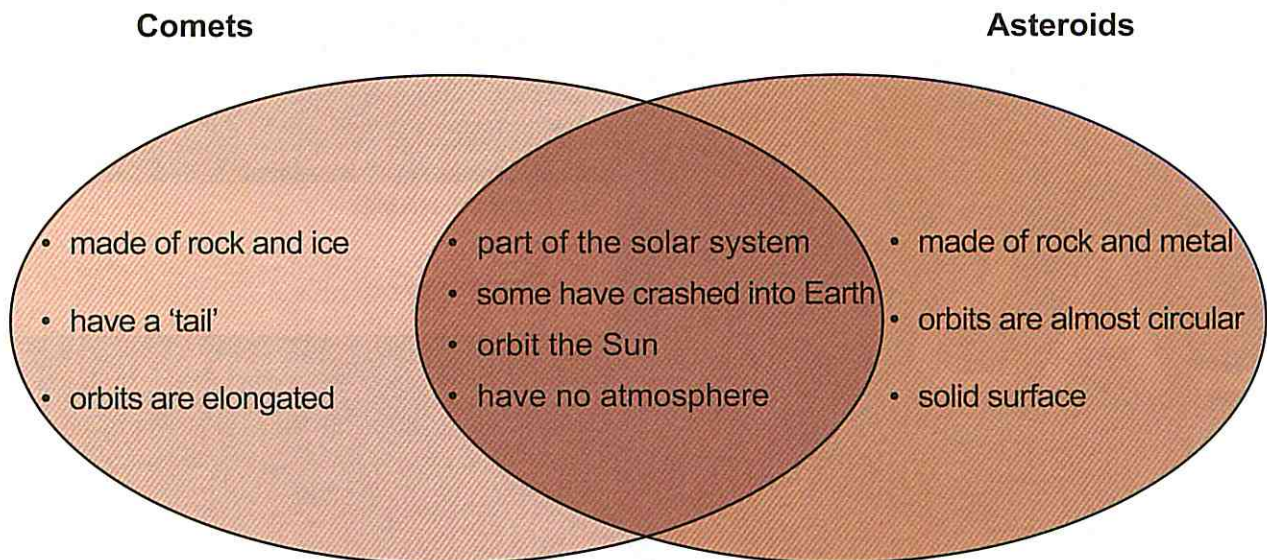


4

Which photographs show bird's feet that are best suited to swimming?

- (A) 1 and 2 (B) 2 and 4 (C) 3 and 4 (D) 1 and 4

2. This diagram lists the features of comets and asteroids.



Which two features do comets and asteroids have in common?

- (A) have a solid surface and circular orbit
(B) have a tail and have an elongated shape
(C) made of rock and ice, and are part of the solar system
(D) have no atmosphere and some have crashed into Earth

3. Plants that have flowers can be classified into two groups called monocots and dicots.

Monocot plants have flowers with three or six petals. Dicot plants have flowers with four or five petals.

These photographs show four flowers.



1



2



3



4

Which flowers are from dicot plants?

- (A) 3 and 4 (B) 2 and 3 (C) 1 and 2 (D) 1 and 3

4. Mary was growing a plant. She measured how tall the plant was at the end of each week, for seven weeks. The plant was 9 cm tall at the start of week 1.

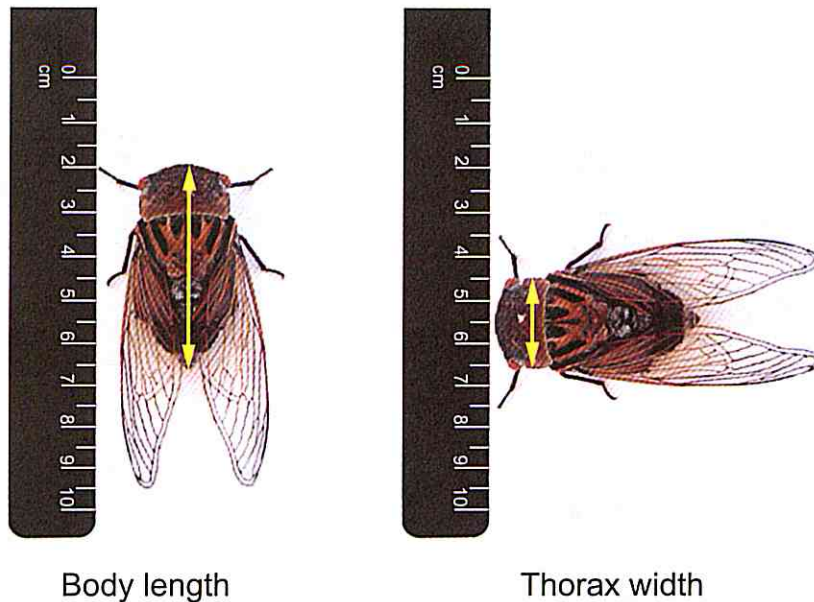
Her results are shown in the table.

End of week	Height of plant (cm)
1	10
2	11
3	14
4	17
5	21
6	23
7	24

During which week did the plant grow the most?

- (A) week 1
(B) week 4
(C) week 5
(D) week 7

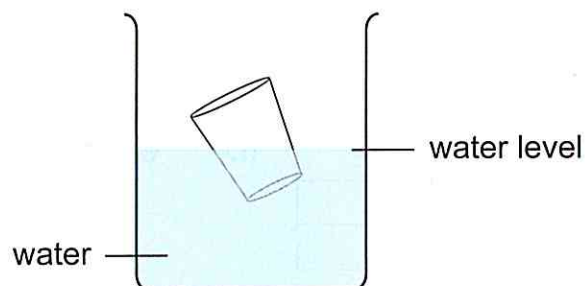
5. In the first photograph the arrow shows the length of a cicada's body.
In the second photograph the arrow shows the width of a cicada's thorax.



Which option has both the correct body length and correct thorax width?

	Body length (cm)	Thorax width (cm)
(A)	$4\frac{1}{2}$	$2\frac{1}{2}$
(B)	$4\frac{1}{2}$	2
(C)	$6\frac{1}{2}$	$2\frac{1}{2}$
(D)	$6\frac{1}{2}$	2

6. Lisa placed an empty plastic cup in a container of water.
The diagram shows the plastic cup floating in the water.



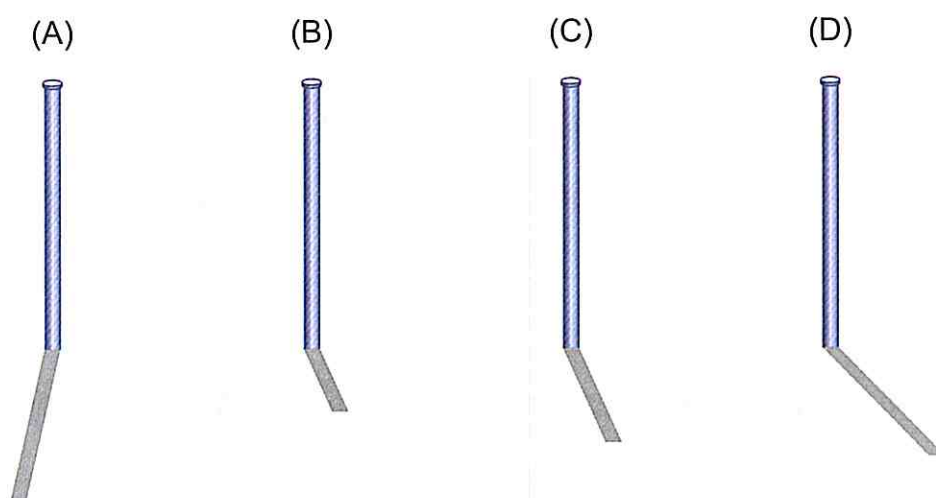
What will happen if Lisa places a marble into the plastic cup?

- (A) The water level will move downwards.
- (B) The water level will move upwards.
- (C) The water level will not change.
- (D) The amount of water in the container will increase.

7. Sara noticed that the shadow made by the flagpole on sunny days changed position throughout the day. The diagrams show the flagpole and its shadow at different times during a day.

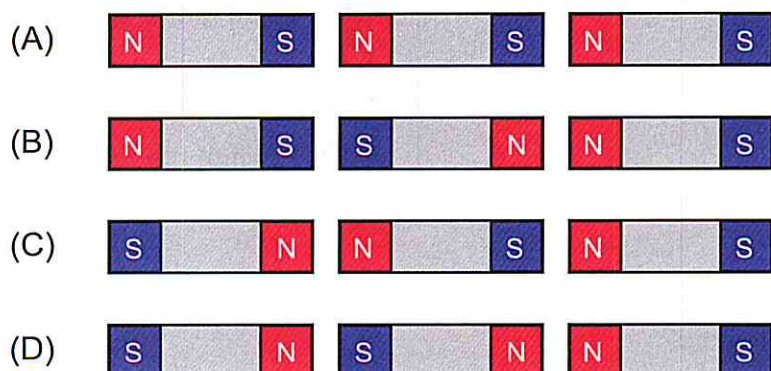


Which diagram shows the shadow of the flagpole at 2 pm on this day?



8. The ends of a magnet are called poles. One end is the north pole (N) and the opposite end is the south pole (S). Two different poles will move towards each other. Two poles that are the same will push away from each other.

In which diagram will all three magnets move towards each other?



For questions 9 and 10 use the information below.

The table shows the fastest speed that some animals can run.

Animal	Fastest speed (km/h)
cheetah	120
emu	48
giraffe	51
greyhound dog	63
human	44
polar bear	30

9. Which animals can run faster than a giraffe?

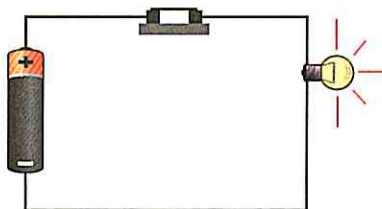
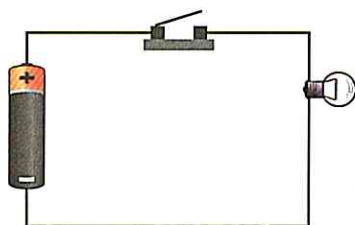
- (A) emu and cheetah (B) greyhound dog and human
(C) cheetah and greyhound dog (D) human and emu

10. How much faster than a human can a cheetah run?

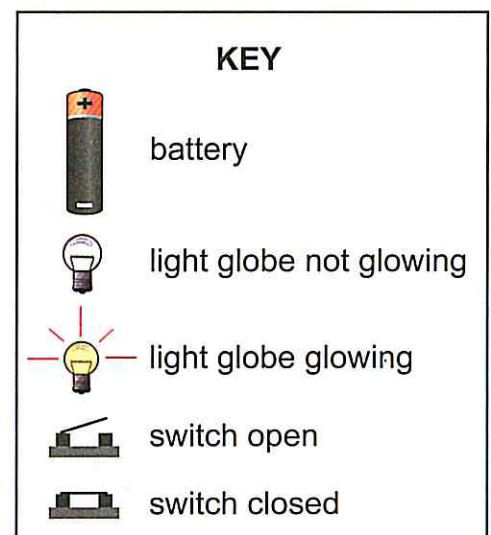
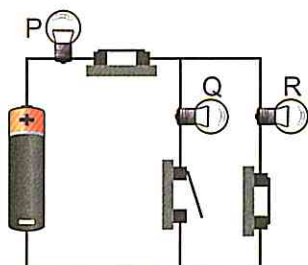
- (A) 120 km/h (B) 86 km/h (C) 76 km/h (D) 44 km/h

11. Sally is investigating electrical circuits.

She uses some diagrams to help her work out what is needed to make a light globe glow.



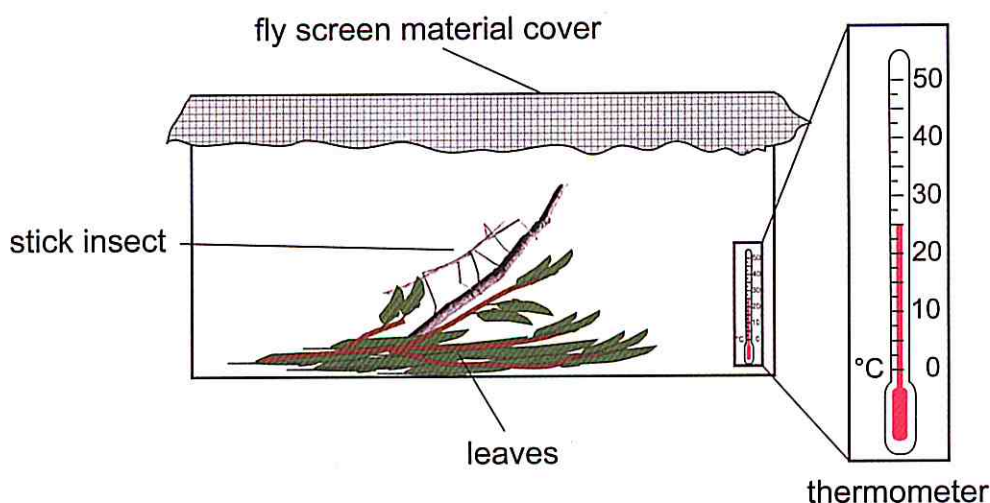
Sally wires up this circuit. The light globes all work and the battery produces enough electricity to make them all glow.



Which light globes in her circuit will glow?

- (A) P and R only (B) P and Q only (C) Q and R only (D) P, Q and R

12. The diagram shows a glass tank set up to keep stick insects. Stick insects only eat leaves.

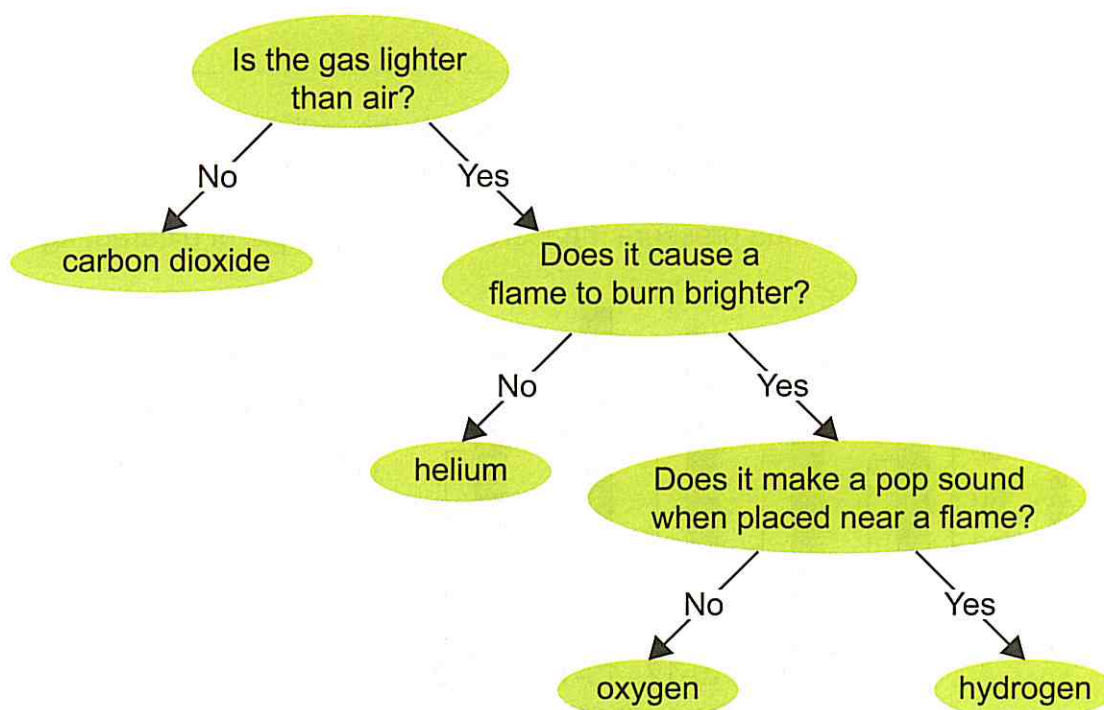


What is the temperature ($^{\circ}\text{C}$) in the tank?

- (A) 20 (B) 22 (C) 25 (D) 30

13. Hannah has four test tubes each containing a different gas. She does not know which test tube contains which gas.

She uses this flow chart to work out which gas is in each test tube.



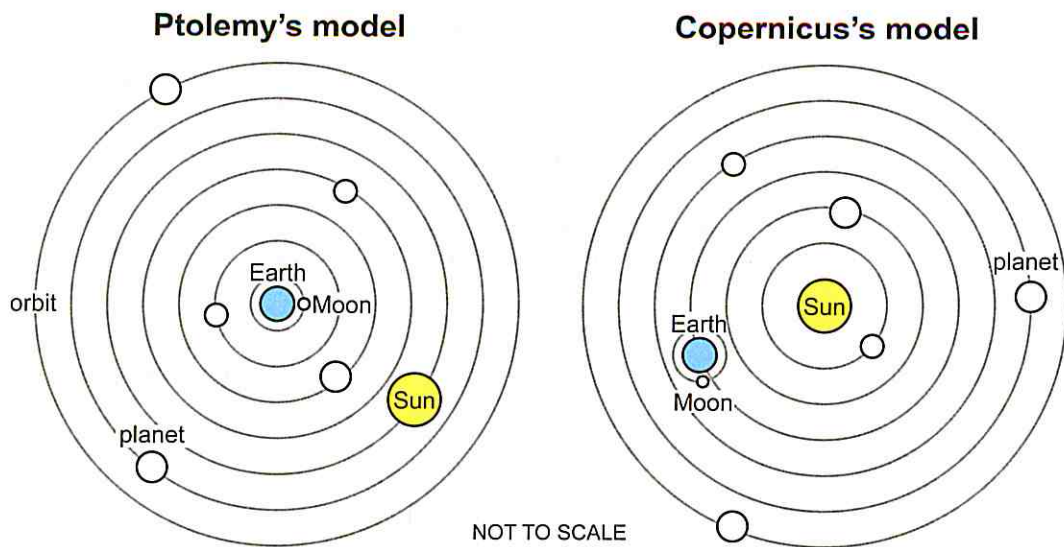
The gas in test tube 1 is lighter than air, makes a lit match burn brighter and makes a loud pop.

Which gas is in test tube 1?

- (A) carbon dioxide (B) helium (C) oxygen (D) hydrogen

14. Ptolemy and Copernicus were astronomers who suggested two different ways in which Earth, the Moon, the other planets and the Sun move.

An orbit represents the path taken by an object.

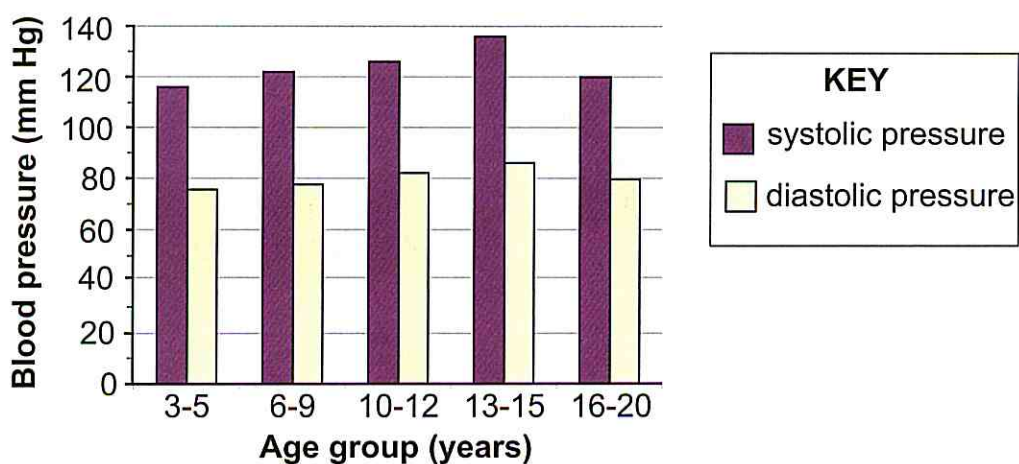


Based on the two models, which statement is correct?

- (A) Ptolemy's model suggests that the Sun is the centre of the universe.
- (B) Copernicus's model suggests that the planets orbit the Sun.
- (C) Ptolemy's model suggests that the Moon orbits the Sun.
- (D) Copernicus's model suggests that the Sun orbits Earth.

15. Doctors check the health of their patients by measuring the patient's blood pressure. They measure two values, the systolic pressure and the diastolic pressure.

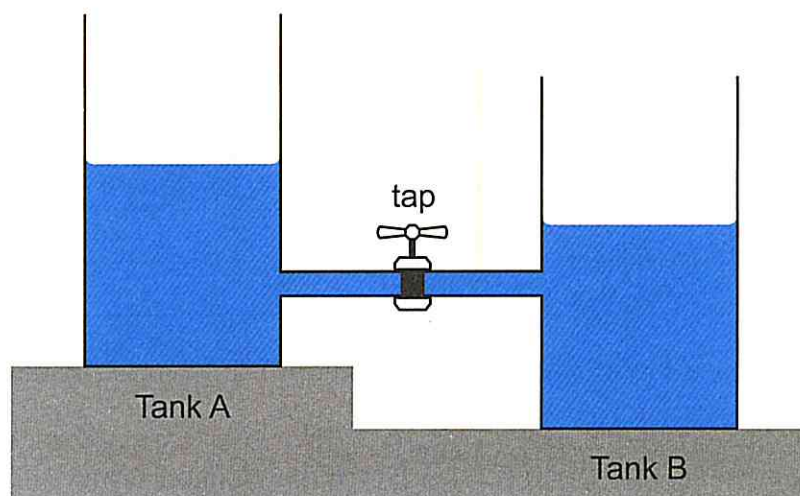
The graph shows normal blood pressure measurements for people of different ages.



What is a normal diastolic blood pressure measurement (mm Hg) for an 8-year-old child?

- (A) 78
- (B) 82
- (C) 116
- (D) 122

16. This diagram shows two tanks connected by a pipe with a tap that controls the flow of water.

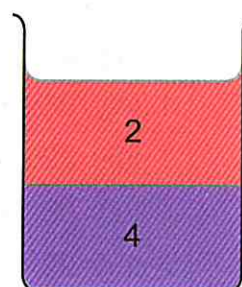
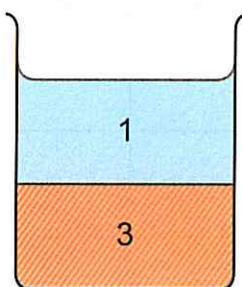
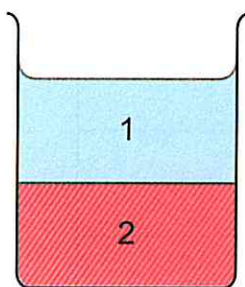


When the tap is opened, what will happen to the water level in each tank?

	Tank A	Tank B
(A)	↑	↑
(B)	↑	↓
(C)	↓	↑
(D)	↓	↓

17. Lily has four different liquids that do not mix with each other. She tested three different combinations of these liquids.

The diagrams show her results.



Which statement is supported by the diagrams?

- (A) Liquid 1 will float on liquid 4.
 (B) Liquid 2 will float on liquid 3.
 (C) Liquid 3 will float on liquid 2.
 (D) Liquid 3 will float on liquid 4.

18. On 1 June, Jacob bought four brands of milk and placed them in his fridge. He tested how long it took each brand of milk to turn sour from its use-by date.

Brand	Use-by date	Date milk turned sour
Cow Valley	3 June	8 June
Milkman's Own	11 June	19 June
Cow Juice	8 June	18 June
Milko Farm	9 June	18 June

Which brand of milk took the longest time to turn sour from its use-by date?

- (A) Cow Valley (B) Milkman's Own
(C) Cow Juice (D) Milko Farm

19. Muli collects eggs laid by her chickens. She collects many eggs each day. Sometimes not all the collected eggs are used.

Muli's mother said, 'Eggs that are old will float in water.'

How can Muli test if this is true?

- (A) Try to float an egg she has in the fridge.
(B) Try to float an egg that was recently laid.
(C) Try to float an egg that was recently laid and an egg from the supermarket.
(D) Try to float an egg that was recently laid and an egg that was laid over a month ago.

20. The table shows the dates and times for high and low tides in a harbour. These times form a pattern.

Date and time of the tide	High tide or low tide
1 July, 01:56 am	low
1 July, 08:01 am	high
1 July, 02:06 pm	low
1 July, 08:11 pm	high
2 July, 02:16 am	low

Based on the information above, what will be the date and time of the next high tide?

- (A) 2 July, 08:21 am (B) 2 July, 02:26 pm
(C) 2 July, 08:11 pm (D) 3 July, 02:26 am

For questions 21 and 22 use the information below.

This table lists the maximum length and mass of different types of adult whales.

Type of whale	Maximum length of adult (metres)	Maximum mass of adult (tonnes)
Arnoux's beaked	9	8
Blue	30	150
Fin	27	90
Humpback	16	48
Minke	10	10
Southern bottlenose	9	4.5
Southern right	17	100
Sperm	15	40

21. Which three types of whales are in order of increasing maximum length?

- (A) Blue, Fin, Southern right (B) Sperm, Humpback, Fin
(C) Sperm, Southern right, Humpback (D) Southern right, Blue, Fin

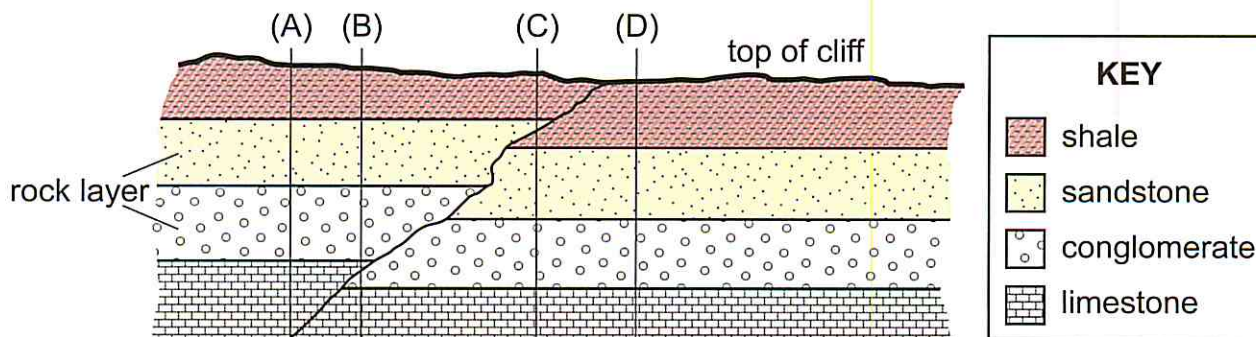
22. Which two types of whales have the closest maximum mass?

- (A) Arnoux's beaked and Southern bottlenose (B) Humpback and Sperm
(C) Blue and Southern right (D) Arnoux's beaked and Minke

23. The diagram shows a cliff face with different layers of rock. A rock climber climbed from the top of the cliff face. She identified the following order of rocks as she moved down the cliff face:

1. shale
2. sandstone
3. conglomerate
4. limestone
5. conglomerate
6. limestone

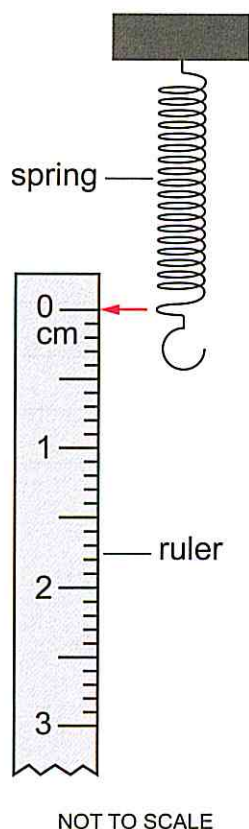
Which path did she take down the cliff face?



For questions 24 and 25 use the information below.

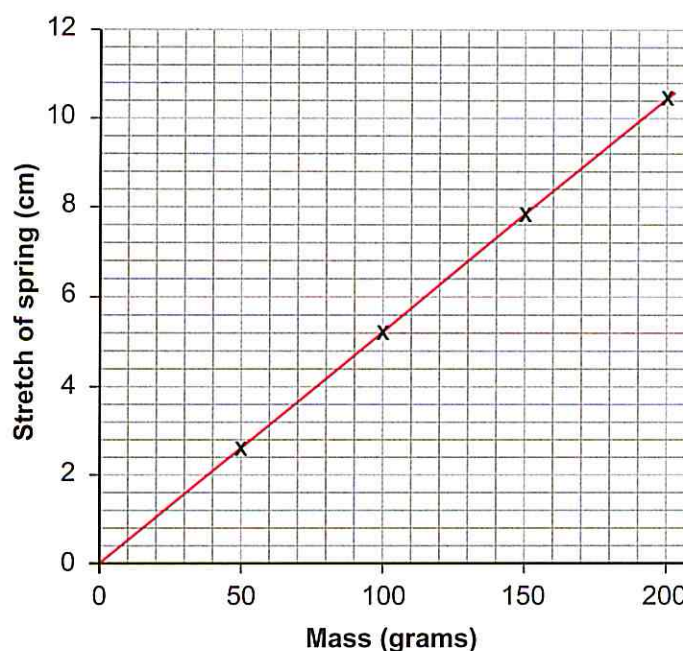
Amy carried out an investigation using a spring and some masses.

She set up her equipment as shown.



She attached a mass to the spring and then measured and recorded how far the spring stretched. She repeated the experiment several times using different masses.

The graph shows her results.



24. Amy removed the mass and replaced it with some keys. The keys made the spring stretch by 7.2 cm.

What was the approximate mass of the keys?

- (A) 145 grams (B) 140 grams (C) 130 grams (D) 125 grams
25. How much would the spring stretch with a mass of 20 grams attached to it?
- (A) 1 cm (B) 2 cm (C) 3 cm (D) 4 cm

26. Sally and Joe are standing on their skateboards, on a smooth flat surface. They are facing in the same direction. Sally and Joe have the same body mass.



Sally gently pushes Joe and they both start moving.

In which directions do Joe and Sally move?

	Sally moves	Joe moves
(A)	forward	forward
(B)	forward	backward
(C)	backward	forward
(D)	backward	backward

27. Different animals have different hearing ranges. An animal's lower limit of hearing is the lowest frequency sound it can hear. Its upper limit of hearing is the highest frequency sound it can hear. The hearing range of an animal is made up of all the sound frequencies in between.

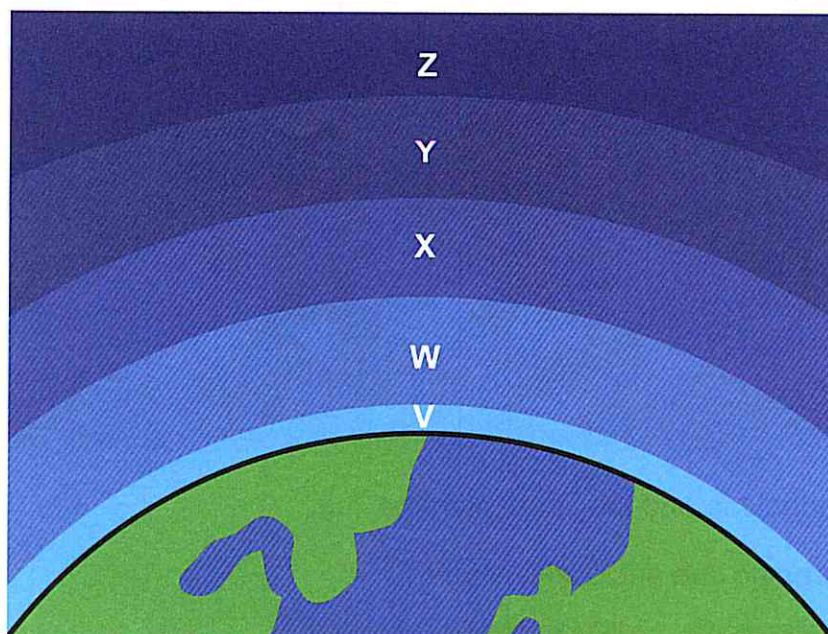
The table shows information about some animals.

Animal	Lower limit of hearing (Hz)	Upper limit of hearing (Hz)
bird	250	21 000
crocodile	20	6 000
dog	15	50 000
bat	1 000	150 000
African moth	5 000	110 000
dolphin	150	150 000

Which animal has the largest hearing range?

- (A) dog (B) bat
(C) African moth (D) dolphin

28. The diagram shows the layers that form Earth's atmosphere.



- The exosphere is the furthest layer from Earth's surface and continues to outer space.
- The troposphere is where most of our weather occurs. Living things are found in this layer.
- The stratosphere is where jet aircraft fly.
- The thermosphere is where satellites orbit Earth, above the flight path of jet aircraft.
- The mesosphere is the middle layer and it is where many meteors burn up.

Which option matches the letter to the layer of the atmosphere?

	V	W	X	Y	Z
(A)	troposphere	stratosphere	mesosphere	thermosphere	exosphere
(B)	troposphere	thermosphere	mesosphere	stratosphere	exosphere
(C)	thermosphere	stratosphere	troposphere	mesosphere	exosphere
(D)	troposphere	mesosphere	exosphere	thermosphere	stratosphere

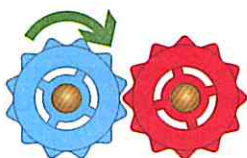
29. Matter can exist as a solid, a liquid or a gas. In a laboratory, substance X melted at 60 °C and boiled at 85 °C.

Which option shows the state of substance X at the temperatures indicated?

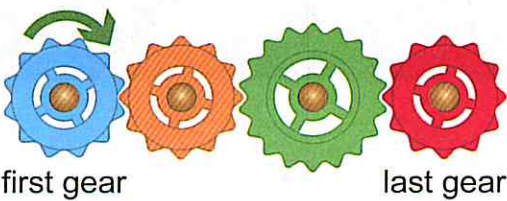
	State of X at 50 °C	State of X at 100 °C
(A)	solid	gas
(B)	solid	liquid
(C)	liquid	gas
(D)	gas	solid

30. Gears are wheels with teeth.

The table shows three sets of gears. In each set, the first gear makes 30 turns per minute in the direction shown. For each set of gears, the table shows the direction of rotation and speed of rotation of the last gear.

	Last gear	
	Direction of rotation	Speed of rotation
 first gear last gear		30 turns per minute
 first gear last gear		20 turns per minute
 first gear last gear		30 turns per minute

This diagram shows another set of gears. The first gear makes 30 turns per minute in the direction shown.



For this set of gears, what are the direction and the speed of rotation of the last gear?

	Direction of rotation	Speed of rotation
(A)		20 turns per minute
(B)		30 turns per minute
(C)		20 turns per minute
(D)		30 turns per minute

Acknowledgment

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Sources

Question 1

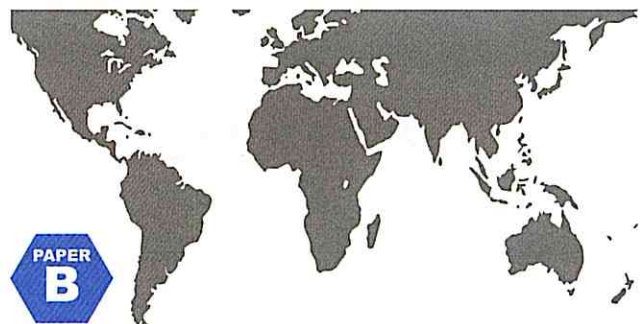
Whooper swan feet image by Aomorikuma <<http://commons.wikimedia.org/wiki/File:Whooper_Swan_feet.JPG>>. Little egret feet image by J.M.Garg <<[http://commons.wikimedia.org/wiki/File:Little_Egret_\(Egretta_garzetta\)_feet_in_AP_W2_IMG_3625.jpg](http://commons.wikimedia.org/wiki/File:Little_Egret_(Egretta_garzetta)_feet_in_AP_W2_IMG_3625.jpg)>>. Steller's sea eagle feet image by 4028mdk09 <<http://commons.wikimedia.org/wiki/File:Riesensee_adler_Fang.JPG>>. American herring gull feet image by Fcb981 <<http://commons.wikimedia.org/wiki/File:Larus_smithsonianus_feet1.jpg>>. Images licensed under the Creative Commons Attribution-ShareAlike 3.0 Unported licence <<<http://creativecommons.org/licenses/by-sa/3.0>>>.

Question 3

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The following year levels should sit THIS Paper:

Australia	Year 4
Brunei	Primary 4
Indonesia	Year 5
Malaysia	Standard 4
New Zealand	Year 5
Pacific Region	Year 5
Singapore	Primary 3
South Africa	Grade 4



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ICAS Paper B Grade 4 Science 2015

Question	Answer
1	A
2	B
3	B
4	A
5	D
6	D
7	D
8	C
9	B
10	C
11	B
12	B
13	A
14	A
15	C
16	A
17	A
18	A
19	B
20	C
21	D
22	B
23	C
24	D
25	D
26	C
27	A
28	C
29	D
30	D